MUHAMMED AFSAL P M

Agentic AI Engineer

Kochi, Kerala, India | Valid Indian Passport

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Kaggle | HuggingFace

PROFILE SUMMARY

Agentic AI Engineer with 3+ years of experience designing and deploying production-grade agentic AI systems, multi-agent orchestration platforms, NL-to-SQL assistants, document intelligence platforms, and computer vision solutions across oil & gas, BFSI, healthcare, e-commerce & affiliate marketing (AdTech), and enterprise automation. Skilled in LangGraph, LangChain, and CrewAI multi-agent workflows, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP) servers, LLM tool calling, and vision-language models (VLMs), working with OpenAI models and locally hosted open-source LLMs (Qwen, Llama) via Ollama. Experienced in building real-time voice and chat AI systems with speech-to-text (STT) and text-to-speech (TTS), FastAPI microservices, vector databases, and SQL/NoSQL databases (PostgreSQL, MySQL, MongoDB), backed by LLMOps practices including Langfuse tracing and evaluation, Drone CI/CD, Docker, and Kubernetes deployments on AWS and Azure. Strong foundation in end-to-end machine learning, including model training, fine-tuning, deployment, and monitoring with a focus on performance, scalability, and reliability.

SKILLS SUMMARY

- **Generative AI & LLMs:** LLMs, Generative AI, Agentic AI, Multi-Agent Systems, RAG, NL-to-SQL, LLM Fine-tuning, LLM Tool Calling, Model Context Protocol (MCP), Human-in-the-Loop (HITL), Prompt Engineering, Embeddings
- **Models & Platforms:** OpenAI (GPT-4), Qwen, Llama, Gemini, Groq, Ollama (Local LLMs), Vision-Language Models (Qwen3-VL, LLaVA), Hugging Face Transformers
- **Frameworks & LLMOps:** LangGraph, LangChain, CrewAI, RAGFlow, Langfuse, Supermemory, OpenAI API, Streamlit
- **ML, Vision & Speech:** Machine Learning, scikit-learn, TensorFlow, Computer Vision, YOLO, DeepSORT, Speech-to-Text (STT), Piper TTS, LiveKit, Twilio
- **Programming & Databases:** Python, SQL, FastAPI, PostgreSQL, MySQL, MongoDB, SQLite, Redis, Vector Databases (ChromaDB, FAISS, Qdrant)
- **Cloud & DevOps:** AWS (S3), Azure, Docker, Docker Compose, Kubernetes, CI/CD (Drone CI, GitHub Actions), Git, RabbitMQ, Celery, SonarQube
- **Other Skills:** Microservices Architecture, REST API Design, API Integration, Model Deployment, Data Engineering, Data Visualization (Power BI, Tableau), Async Programming

WORK EXPERIENCE

NGXP Technologies LLP *Jun 2026 – Present*

Agentic AI Engineer *Kochi, Kerala, India*

- Built an NL-to-SQL database assistant for an oil & gas client using LangGraph multi-agent orchestration, locally hosted LLMs (Qwen via Ollama), and OpenAI models, translating natural language business questions into accurate SQL queries over enterprise databases, with voice interaction through speech-to-text (STT) and Piper TTS.
- Developed AI-driven features for oil & gas operations, including crew management, well slippage tracking, automated daily report generation, and data anomaly detection for well monitoring, along with a database migration validator that automates data validation checks and migration reporting.
- Architected a multi-agent orchestrator platform using LangGraph, where a supervisor agent plans tasks across specialized agents (web search, communication, verifier), executes steps in parallel, re-plans on failure, and routes sensitive actions through human-in-the-loop approvals with code-enforced policy guardrails, containerized with Docker Compose.

- Built an enterprise document intelligence platform using RAGFlow, vector databases, and local LLMs (Qwen 35B via Ollama) alongside OpenAI models, implementing LLMOps with Langfuse for tracing and evaluation, Supermemory for persistent agent memory, and Drone CI/CD pipelines for automated deployment.
- Developed a real-time driver safety monitoring system combining on-device detection models with vision-language models (Qwen3-VL 8B, LLaVA 7B) on self-hosted Ollama to verify phone use, smoking, eating, drinking, and seatbelt violations through a FastAPI service, reducing false alerts with structured JSON outputs and confidence-tiered alert routing.

Parity Cube Private Limited *Apr 2026 – May 2026*
Integration & Automation Engineer *Mumbai, Maharashtra, India*

- Designed and developed a Model Context Protocol (MCP) server for Cuelinks, the company's affiliate marketing platform, exposing REST API endpoints as LLM-accessible tools to enable natural-language interactions through AI clients such as ChatGPT and Claude.
- Engineered LLM-driven tool-calling and API orchestration workflows using endpoint-specific prompts and tool schemas, enabling LLMs to interpret user queries, select appropriate API operations, and generate structured request payloads.
- Developed an agentic competitor intelligence platform using LangGraph, FastAPI, OpenAI, and Docker, integrating external affiliate network APIs to enable cross-platform product discovery and comparison of product availability, offers, and pricing.

Techvantage Systems Pvt. Ltd *Dec 2024 – Apr 2026*
Associate Software Engineer (AI) *Trivandrum, Kerala, India*

- Designed and deployed multi-agent AI systems using CrewAI and LLMs (GPT-4, Groq, Llama), including LLM fine-tuning, to automate verification, risk assessment, and decision-making in banking, insurance, and audit domains.
- Engineered end-to-end ML pipelines following the FTI architecture with scikit-learn and MongoDB, automating feature extraction, model training, and inference with integrated monitoring.
- Built enterprise AI platforms for recruitment, contract intelligence, and HR policy, leveraging adaptive testing and LLM question generation to deliver real-time analytics and ensure compliance.
- Developed advanced computer vision and data retrieval systems using YOLO, DeepSORT, and ChromaDB, enabling automated document analysis, object tracking, and contextual report generation.
- Delivered production-grade solutions by integrating FastAPI, AWS, and SQL/NoSQL databases, implementing features like secure OTP authentication, multi-language RAG, fraud detection, and automated report generation.

Go-Do *(Startup incubated within Techvantage)* *Dec 2025 – Feb 2026*
Core Team Member – Associate Software Engineer (AI) *Trivandrum, Kerala, India*

- Developed a multilingual agentic AI concierge accessible via WhatsApp and voice interfaces, enabling users to complete real-world tasks through natural language commands across multiple Indian languages.
- Architected end-to-end conversational pipelines using FastAPI, LiveKit, Twilio WhatsApp API, Redis, MongoDB, and FAISS-based RAG, supporting real-time task orchestration, session persistence, and knowledge retrieval.
- Implemented multimodal AI workflows integrating STT, TTS, and LLM reasoning (Gemini, OpenAI, Sarvam), delivering scalable voice-and-text agents with tool-calling, analytics logging, and production-ready deployment.

Zentis AI *(Startup incubated within Techvantage)* *May 2025 – Oct 2025*
Core Team Member – Associate Software Engineer (AI) *Trivandrum, Kerala, India*

- Architected and developed a modular microservices-based AI platform from inception using autonomous agents, enabling workflow automation, document intelligence, and decision support

across BFSI, healthcare, and manufacturing domains.

- Implemented AI orchestration pipelines using LLMs (Azure OpenAI, Groq), Retrieval-Augmented Generation (RAG), vector search, and CrewAI multi-agent frameworks to support contextual reasoning and knowledge-driven automation.
- Built scalable backend services with FastAPI, MongoDB, and RabbitMQ; containerized with Docker and deployed on Kubernetes across Azure and AWS with secure APIs, fault tolerance, and distributed processing.
- Established production DevOps practices including CI/CD pipelines, automated testing, SonarQube code quality checks, monitoring, and compliance controls, improving system reliability, observability, and reducing manual audit effort.

Bharat Petroleum Corporation Limited (Kochi Refinery) *Dec 2023 – Dec 2024*
AI Engineer (Apprenticeship) *Kochi, Kerala, India*

- Extracted, validated, and managed data from multiple sources using SQL/MySQL, ensuring high data quality and integrity for petroleum operations; enhanced workflows using agent-based AI for automated data validation and anomaly detection.
- Conducted advanced data analysis using SQL, stochastic methods, and statistical modeling to optimize extraction, refining, and distribution processes, incorporating AI-driven insights for improved decision-making.
- Developed and optimized machine learning models to support AI-powered solutions, including an AI chatbot integrated with a Retrieval-Augmented Generation (RAG) pipeline for context-aware responses.
- Designed and implemented AI-driven data auditing and report generation systems using agent-based architectures and RAG to automate validation, summarization, and compliance reporting.
- Built interactive dashboards with Power BI, Tableau, and Excel to enhance operational efficiency and data-driven decision-making, integrating AI-generated insights and predictive analytics.

PROJECTS

- **Oil & Gas Operations Intelligence Platform:** Built an AI platform with an NL-to-SQL assistant using LangGraph, local LLMs (Qwen via Ollama), and OpenAI models, with STT/Piper TTS voice interaction, well monitoring anomaly detection, crew management, daily reporting, and Langfuse tracing.
- **Document Intelligence Platform:** Built an enterprise RAG platform using RAGFlow, vector databases, and local LLMs (Qwen 35B via Ollama) alongside OpenAI models for document parsing and question answering, with Langfuse tracing and evaluation.
- **Publisher Website Classifier – LLM-Powered Publisher Vetting System:** Built an LLM-powered publisher vetting microservice with RAG (ChromaDB + sentence-transformers), dual-provider architecture (OpenAI/Ollama), and tiered web scraping to automate affiliate quality assessment.
- **Contract Management Tool:** Developed an AI-powered contract analysis and management system using FastAPI, LLMs (Groq/Local), ChromaDB, and MongoDB for automated compliance checks, risk assessment, clause detection, and intelligent contract Q&A chatbot with multi-format document support.
- **AI Agents for Banking Loan Processing:** Developed AI agents to automate loan verification, eligibility checks, and decision-making workflows.
- **AI-Powered Recruitment Platform:** Built recruitment automation integrated with MongoDB for resume shortlisting, test scheduling, AI-generated questions, and candidate analytics.
- **AI Chatbot:** Developed an AI-powered chatbot for HR and operational workflows using agent-based intent classification and LLMs for contextual responses.

EDUCATION

Certified Specialist in Data Science and Analytics *May 2024 – Oct 2024*
ICT Academy of Kerala

Bachelor of Technology in Computer Science and Engineering *Jul 2019 – Aug 2023*
with minor in Electronics and Communication Engineering
First Class with Distinction | CGPA: 8.46
APJ Abdul Kalam Technological University

CERTIFICATIONS & PUBLICATION

- Microsoft Certified: Azure Data Scientist Associate (DP-100) Certificate *Jun 2024*
- Microsoft Certified: Azure Data Fundamentals (DP-900) Certificate *Mar 2024*
- IBM Data Science Professional Certificate Certificate *Dec 2024*
- Introducing Multimodal Llama 3.2 Certificate *Jan 2025*
- Machine Learning Fundamentals Micro-Credential by Alteryx Certificate *Apr 2024*
- Generative AI Fundamentals by Databricks Certificate *Oct 2025*
- **Publication:** "Smart Application for Visually Impaired People," IJARIIE, Vol-9, Issue-3, 2023, ISSN(O) 2395-4396 Publication *Jun 2023*